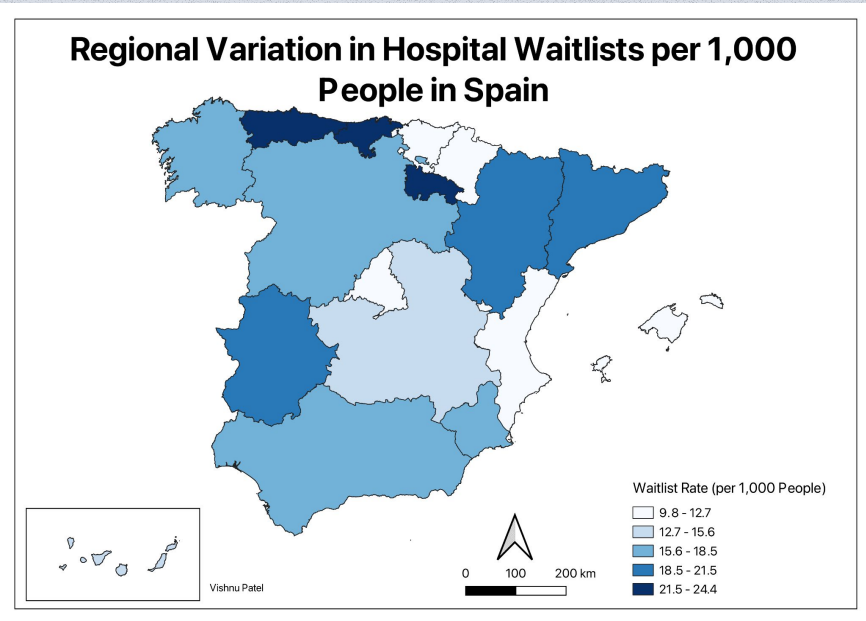


Regional Disparities in Spanish Health Care

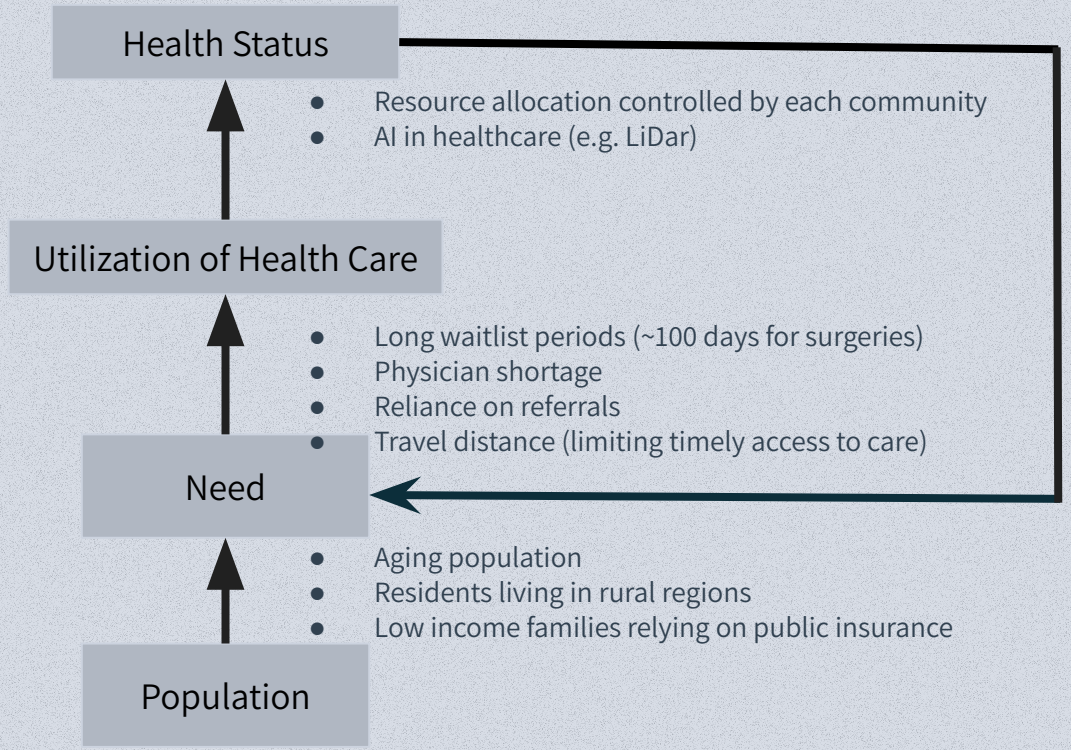
Vishnu Patel

Hospital Waitlists Across Spain

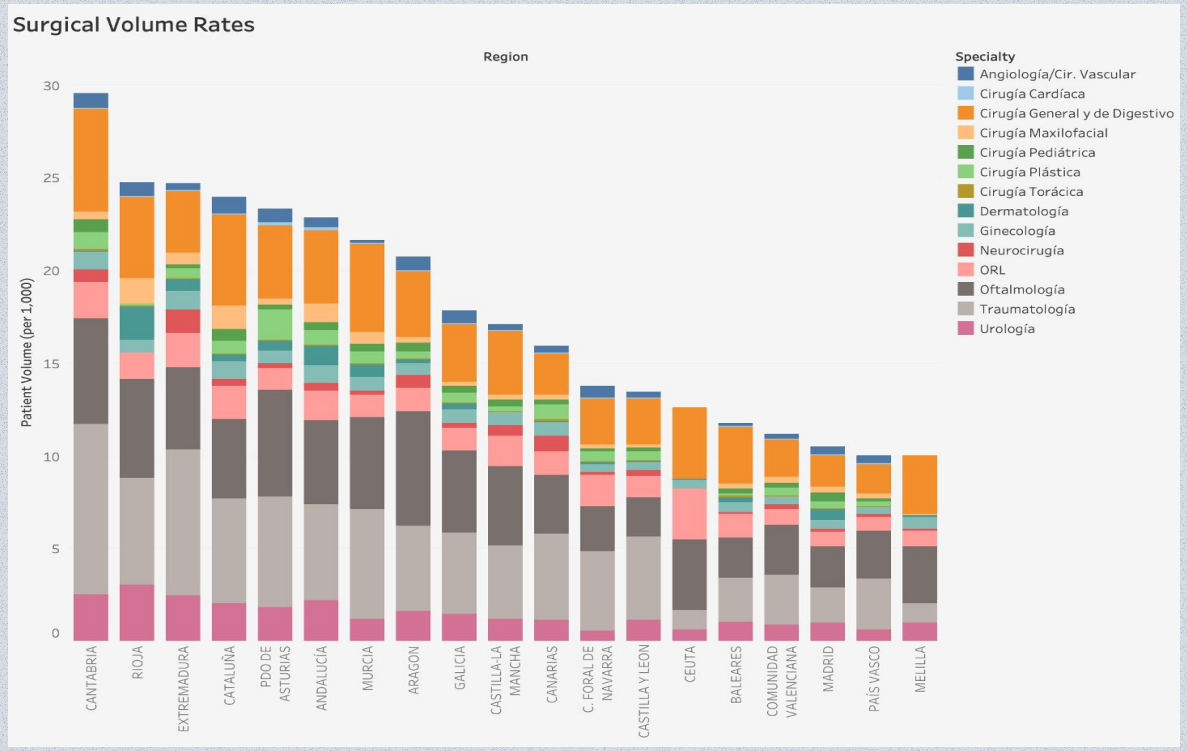


- Choropleth map using GADM boundaries + Statista data
- Normalized waitlists reveal clear regional disparities
- Longest: Castilla-La Mancha, Extremadura
- Shortest: Basque Country, Navarre, Madrid
- Patterns not explained by population alone

Framework: Oleske Model



Regional Access Disparities



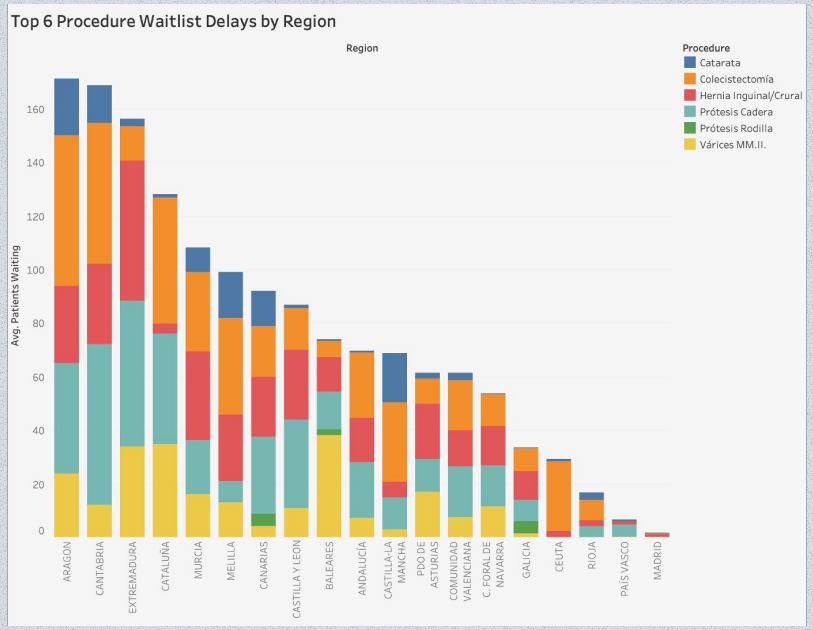
Surgical Bottlenecks



- Plastic Surgery shows severe delays (>400 days in Cantabria and Andalucía); Neurosurgery spikes in Aragon.
- Extremadura, Andalucía, and Cantabria show widespread long wait times, while Madrid, La Rioja, and País Vasco remain consistently efficient (<60 days).

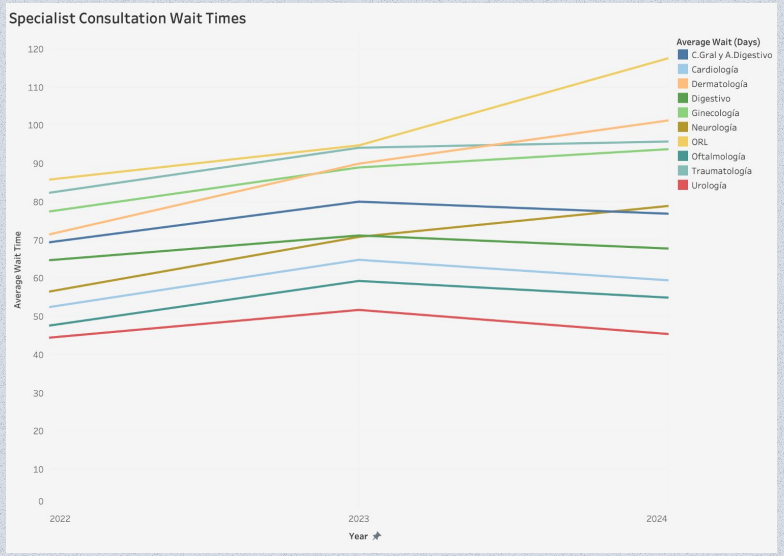
Procedures Driving Demand

- Hip replacements represent the largest waiting block across high-volume surgeries.
- Aragon and Cantabria lead with over 160 average patients waiting per selected procedure, compared to fewer than 10 in Madrid and País Vasco.



Outpatient Access Gaps

- Otorhinolaryngology wait times surged sharply from 86 to nearly 118 days between 2022 and 2024.
- Most specialties peaked in wait days in 2023 and slightly plateaued/improved in 2024, though Traumatology and Dermatology remain high.



Key Takeaways

- **Public-Private Partnerships (PPPs)**: Regions like Madrid collaborate with private facilities to expand capacity; rural regions rely mainly on public providers.
- **Budget Priorities:** Some communities invest in specialty care and surgical capacity, while others prioritize primary care or operate with limited budgets.
- **Geographic Access Imbalances:** Regional location appears more relevant to differences in waiting times than population or regional size alone.
- **Inconsistent Progress:** Improvements in certain specialties have not translated into consistent reductions in waiting times across all regions and services.

Future Research

Quantitative approach

- **Predictive Analytics & AI:** Explore AI-driven models to forecast peak patient demand, optimize staff allocation, and identify potential bottlenecks in surgical scheduling.
- **Socioeconomic Factors:** Expand the data pipeline to incorporate income, employment, and demographic indicators to examine how socioeconomic conditions influence healthcare demand and resource allocation.
- **Supply vs. Demand:** Compare waiting lists with physician availability, hospital capacity, and procedure volumes to identify resource shortages and opportunities for redistribution.

Qualitative approach

- **International Comparisons:** Compare Spain's decentralized healthcare system with systems in Germany and Switzerland to understand how different organizational structures influence access and efficiency.
- **Digital Healthcare Accessibility:** Investigate differences in telehealth adoption, particularly how digital literacy and broadband availability affect access in rural communities.
- **Policy Effectiveness:** Evaluate regional healthcare strategies and identify policies or practices that could help reduce persistent waiting times.